

VISION Resistivity
Dual Frequency
Recorded Mode

Schlumberger

Country: Norway

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

Latitude: 60° 34' 35.14" N

Longitude: 3° 26' 36.13" E

Block: 31/5

UWID:

Rig Name:

Rig Type:

West Hercules

Semi-Submersible

FL: Norwegian North Sea

FL1: Section: 12 1/4 in.

FL2: Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level

Ground Level: 307.00 m

Acquisition Dates: 23-Dec-2019 -- 25-Dec-2019

Log Interval: 1856.00(m) -- 2584.00(m)

Index Types: Measured Depth

Index Scales: 1:200

Depth Source: Driller's Depth

Depth Sensor: 3rd Party Depth

Print Type: Final

Spud Date: 02-Dec-2019

Other Services:

Direction & Inclination (Surveys)

Directional Drilling

Annular Pressure While Drilling

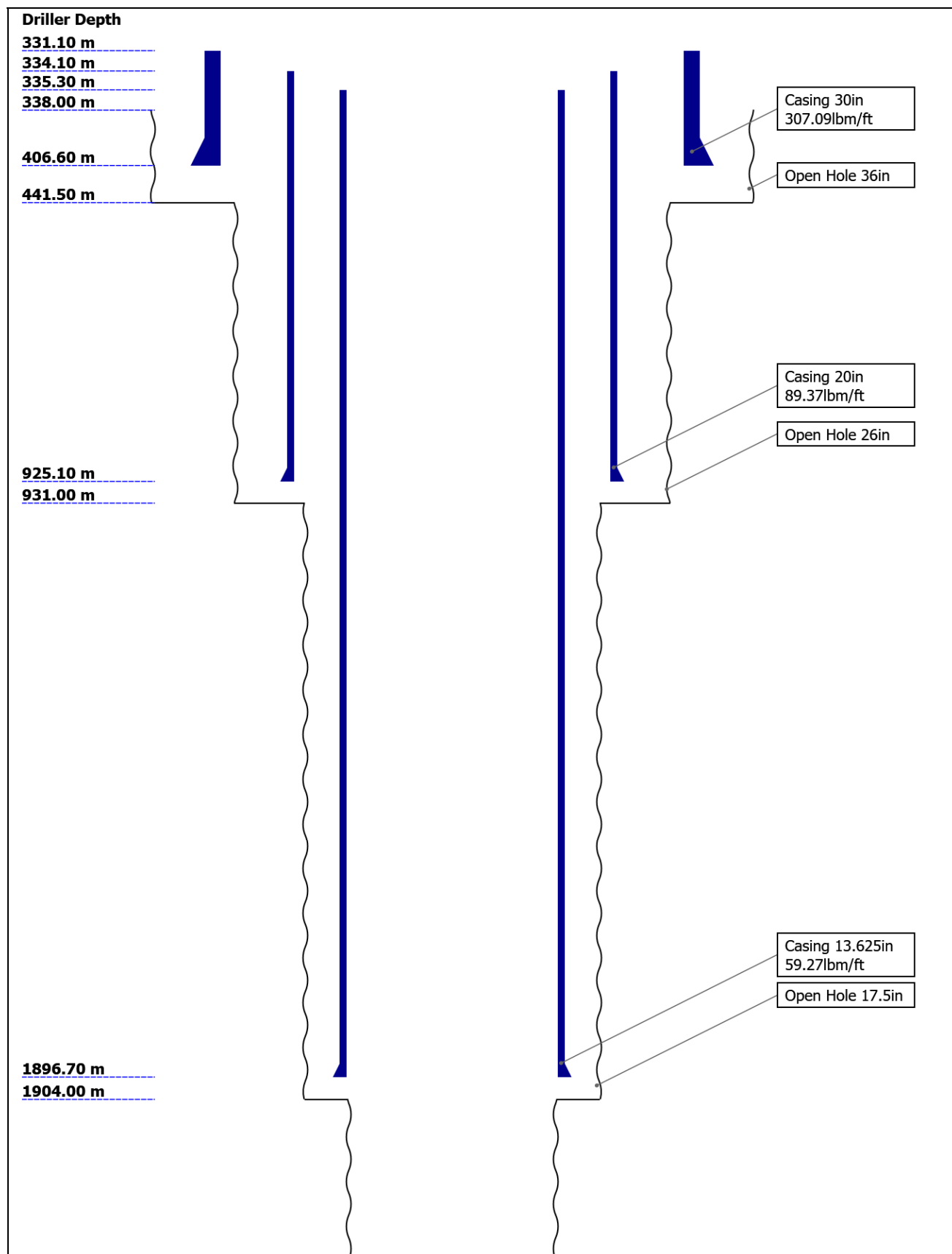
Disclaimer

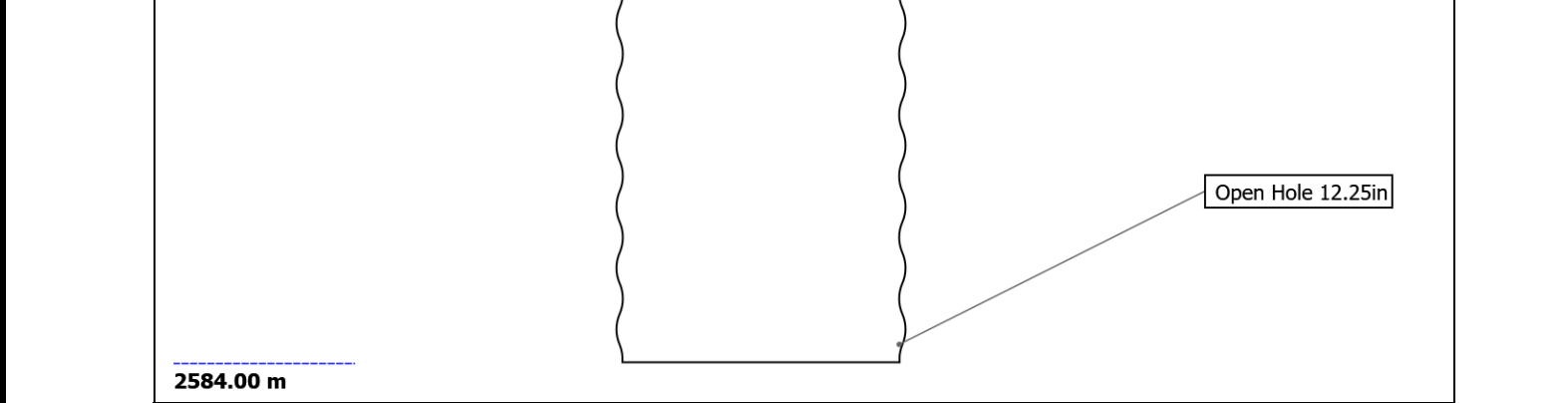
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Well Sketch





Borehole Size/Casing Record

Bit						
Bit Size (in)	36	26	17.5	12.25		
Top Driller (m)	338	441.5	931	1904		
Bottom Driller (m)	441.5	931	1904	2584		
Casing						
Size (in)	30	20	13.625			
Weight (lbm/ft)	307.09	89.37	59.27			
Inner Diameter (in)	28.063	19.164	12.804			
Grade	X56	N80	P110			
Top Driller (m)	331.1	334.1	335.3			
Bottom Driller (m)	406.6	925.1	1896.7			

Operational Run Summary

Parameter (unit)	Run 8					
Date Log Started	23-Dec-2019					
Time Log Started	06:55:46					
Date Log Finished	25-Dec-2019					
Time Log Finished	13:49:45					
Bit Size (in)	12.250					
Bit Start Depth (m)	1904.00					
Bit Stop Depth (m)	2584.00					
Top Log Interval (m)	1900.20					
Bottom Log Interval (m)	2574.80					
Max Hole Deviation (deg)	0.26					
Azimuth of Max Deviation (deg)	22.82					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	G.Boyd					
Witnessed By	J.Eikenes					
Service Order Number	19SCA0085					

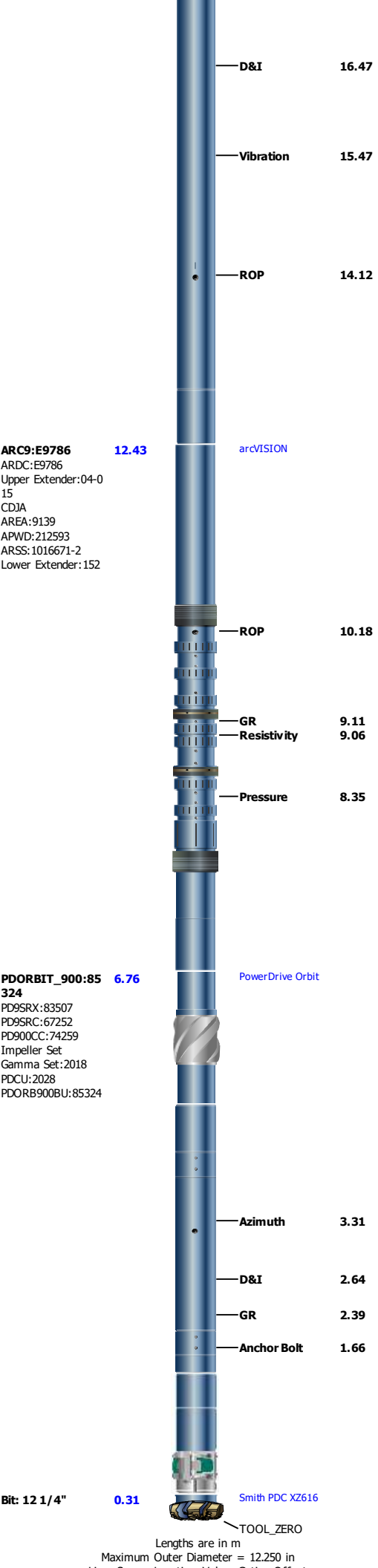
Borehole Fluids

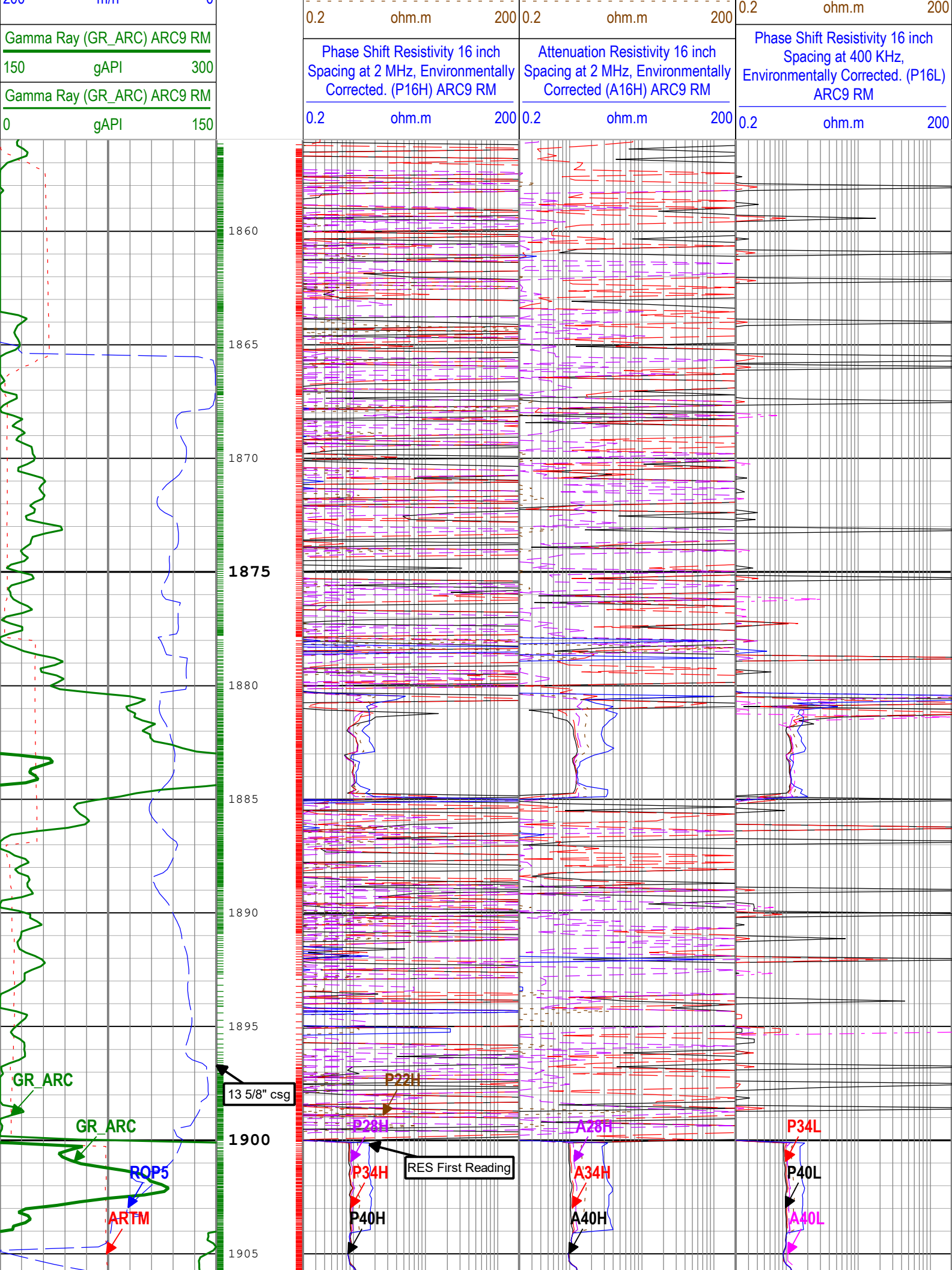
Parameter(unit)	Run 8					
Fluid Type	Water					
Fluid Name	Glydril					
Max Recorded Temperatures (degC)	45					
Source of Sample	Active Tank					
Salinity (ppm)	Zoned					
Density (g/cm3)	Zoned					
Funnel Viscosity (s)						
Fluid Loss (cm3)						
PH						
Source RMF						
RMC	Pressed					
RM @ Meas Temp (ohm.m@degC)	0.07 @ 17.1					
RMF @ Meas Temp (ohm.m@degC)	0.06 @ 18.06					
RMC @ Meas Temp (ohm.m@degC)	0.1 @ 20					
RM @ BHT (ohm.m@degC)	0.04 @ 45					
RMF @ BHT (ohm.m@degC)	0.04 @ 45					
RMC @ BHT (ohm.m@degC)	0.06 @ 45					
Total Solid (%)						
High Gravity Solids (%)						

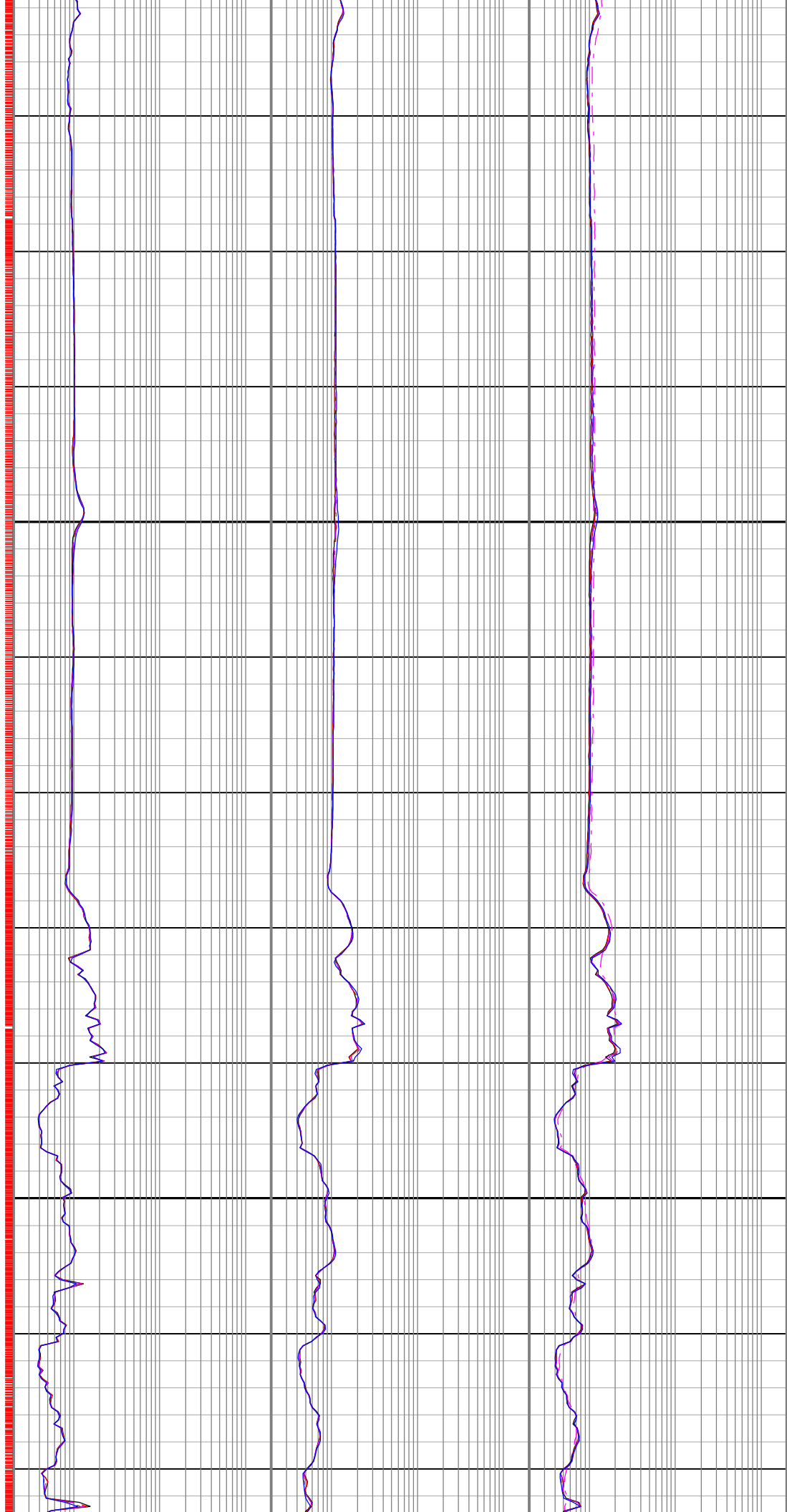
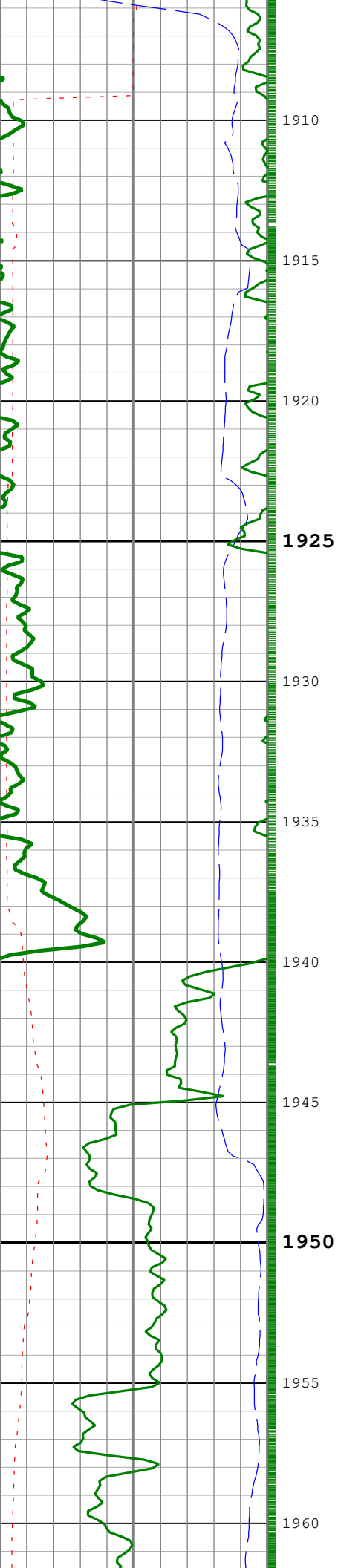
Run 8

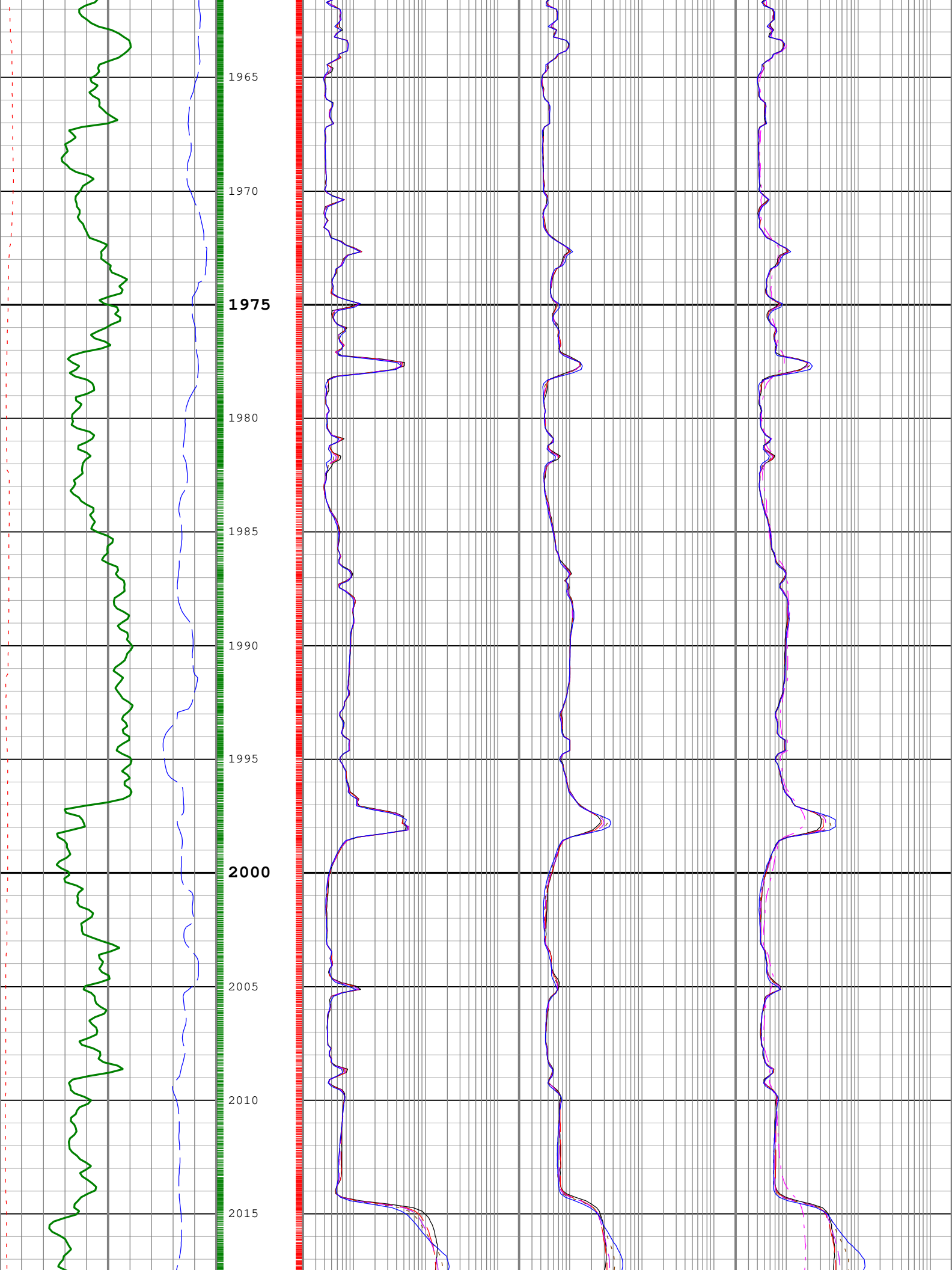
Remarks and Equipment Summary

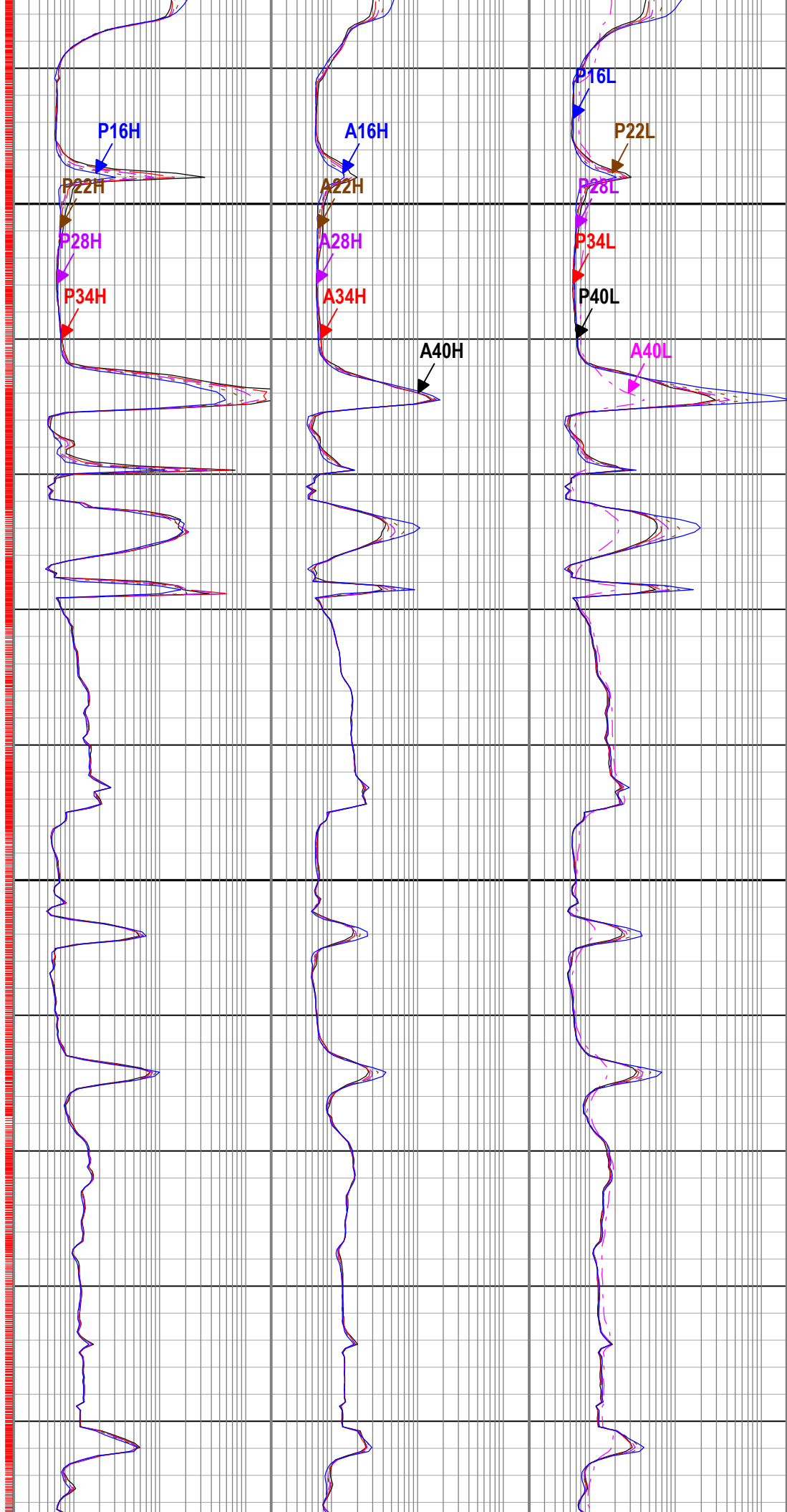
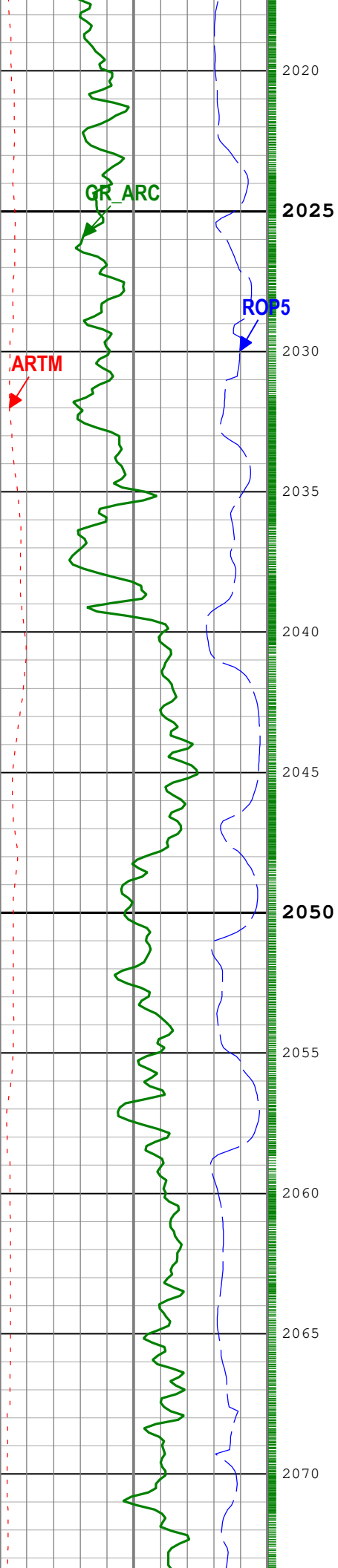
Run 8: Toolstring		Run 8: Remarks	
Equip name NM Stab: 9 1/2":M WS 3732	Length 24.15	MP name 12 1/8 in. Non-Mag Stab	All depths referenced to driller's depth. See EOWR for depth tracking details.
			All data from tool memory. All data acquired while drilling
			Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.
			Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.
			Run 8 and 12 1/4 in. section TD at 2584 m.
X/O: 9 1/2":XOS-947	21.65	X-Over	
TELE900:E9186-IW	20.82	TeleScope	
MSSU900:74066 Upper Extender:02-06-19 MDC900:E9186-IW MMA:7002 MDI:3770 PMVC:43 PMEA:3792 MTA:3757 MTK900:S-298895-7775 MSSD900:888081-1 Lower Extender, Long:10708-9			

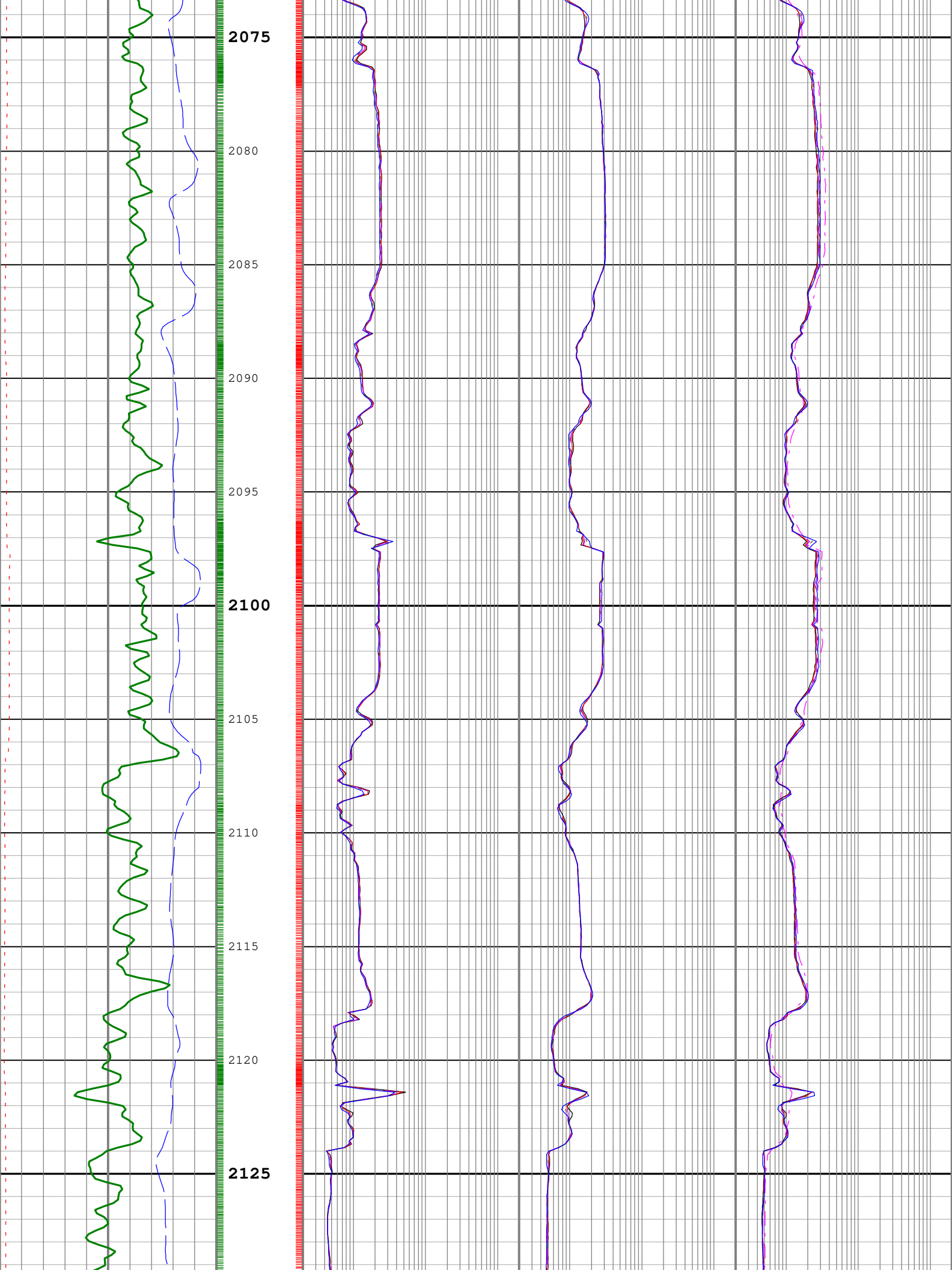


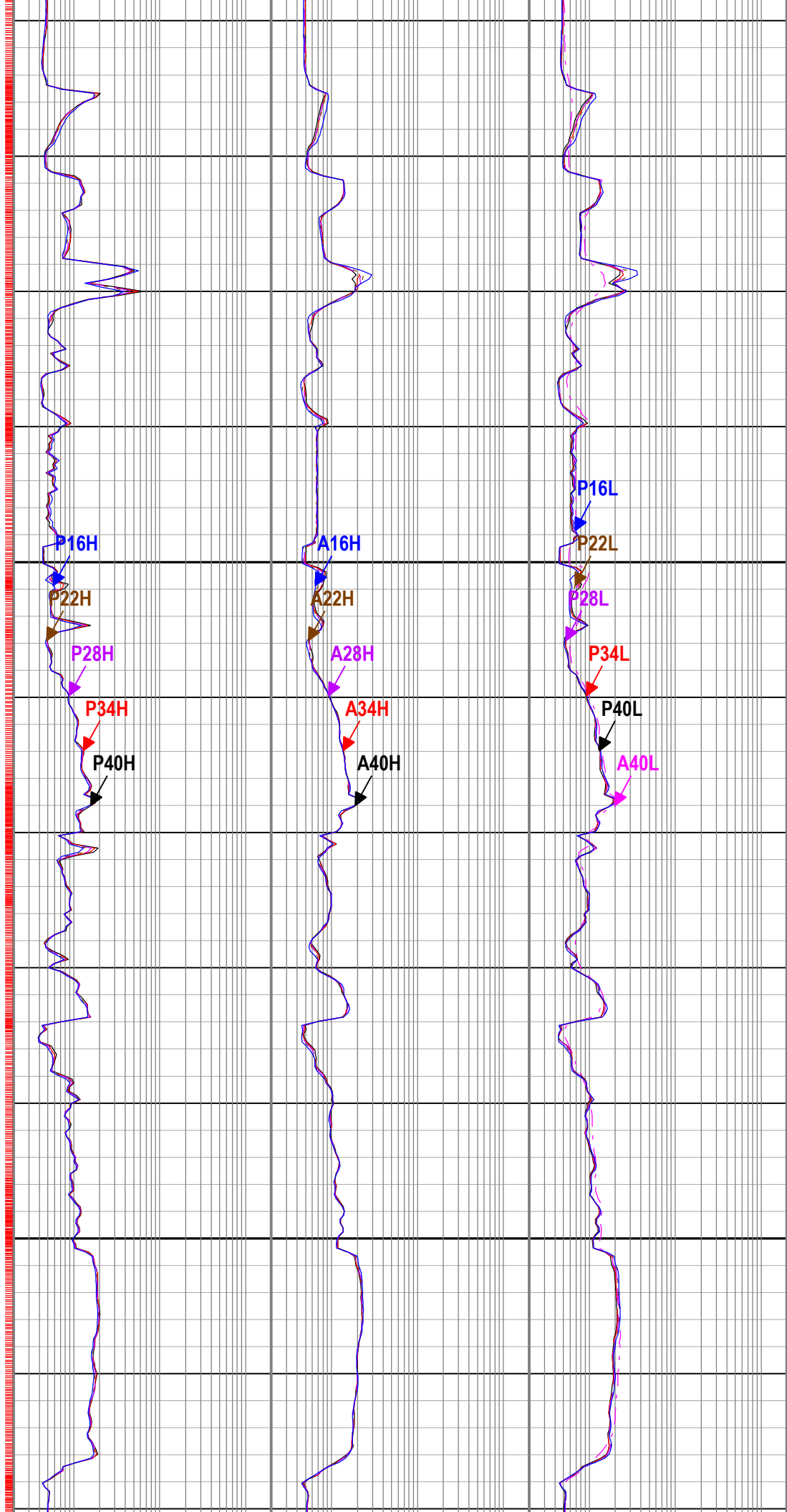
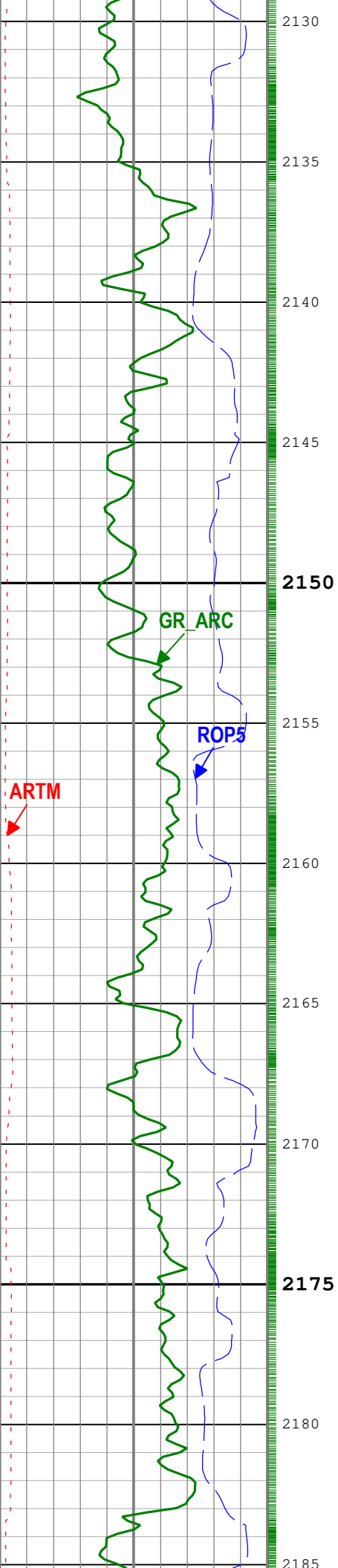


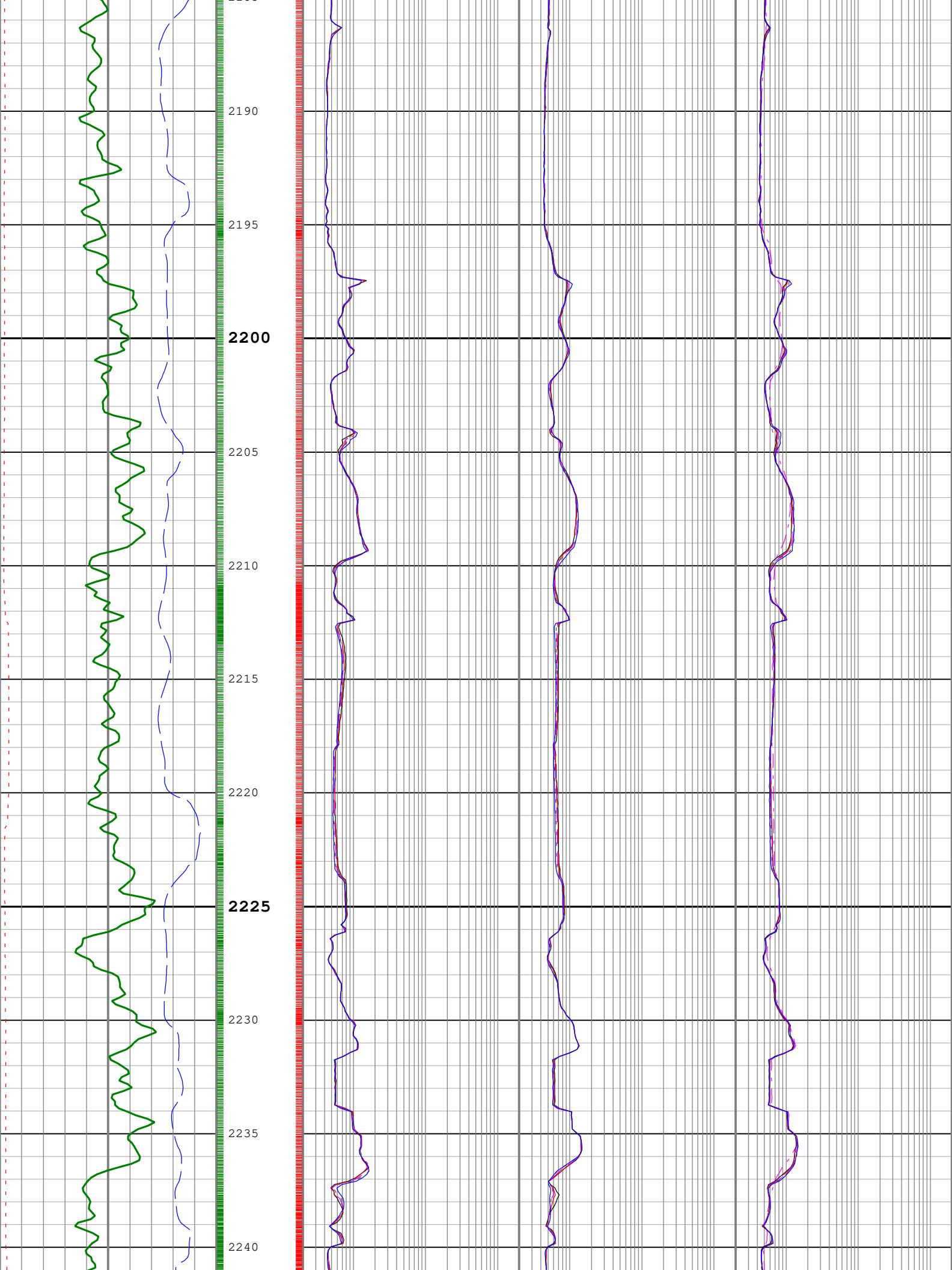


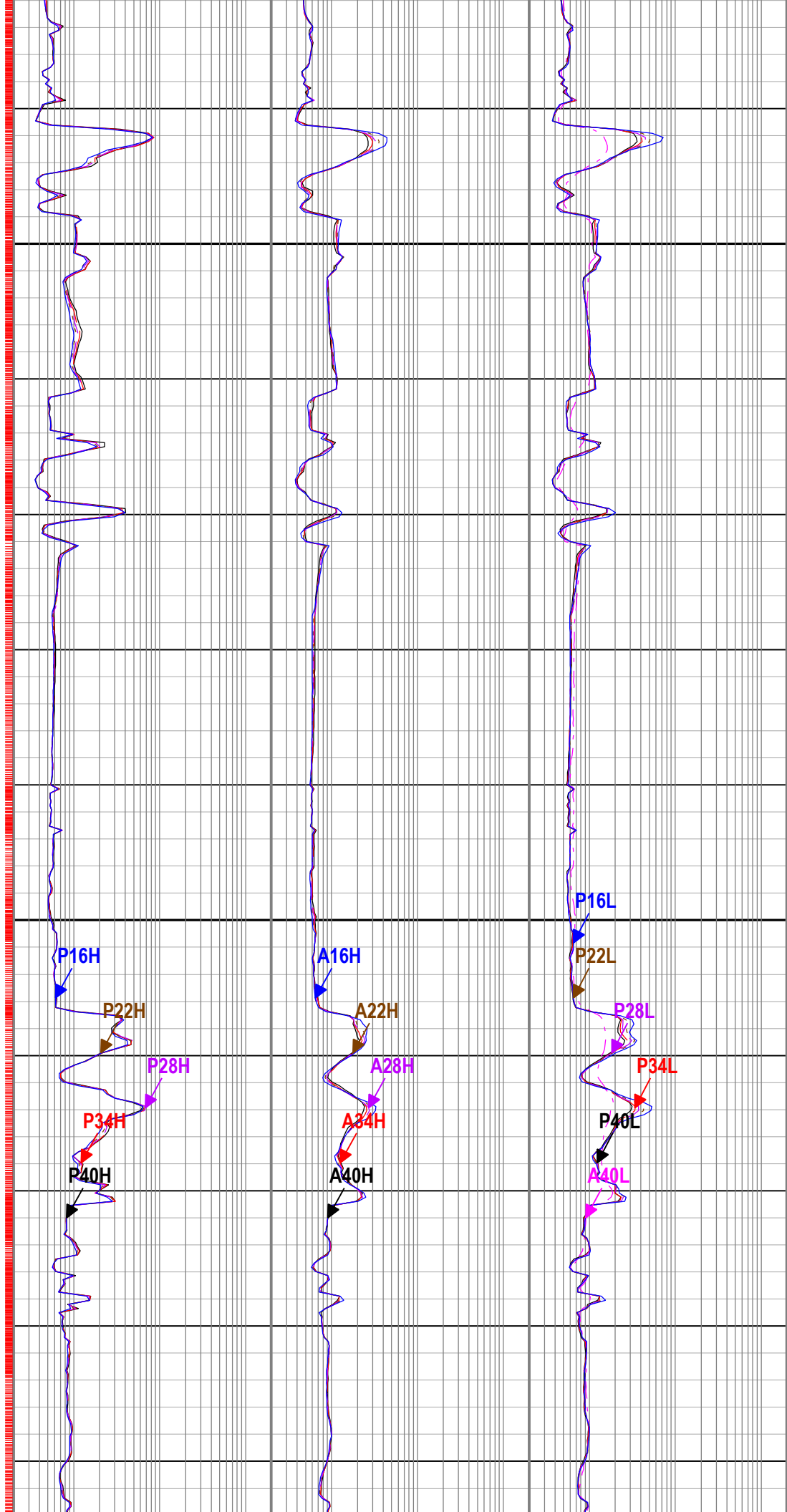
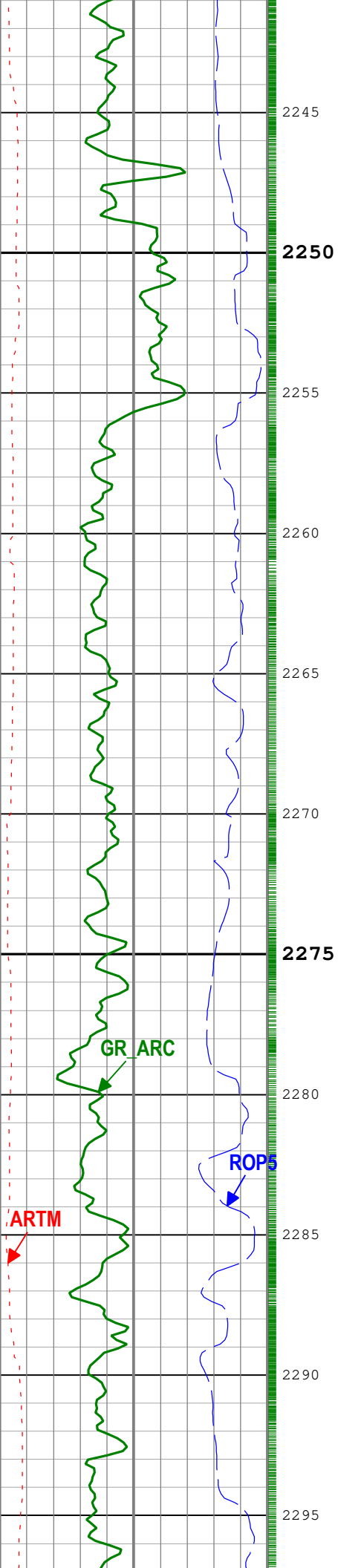




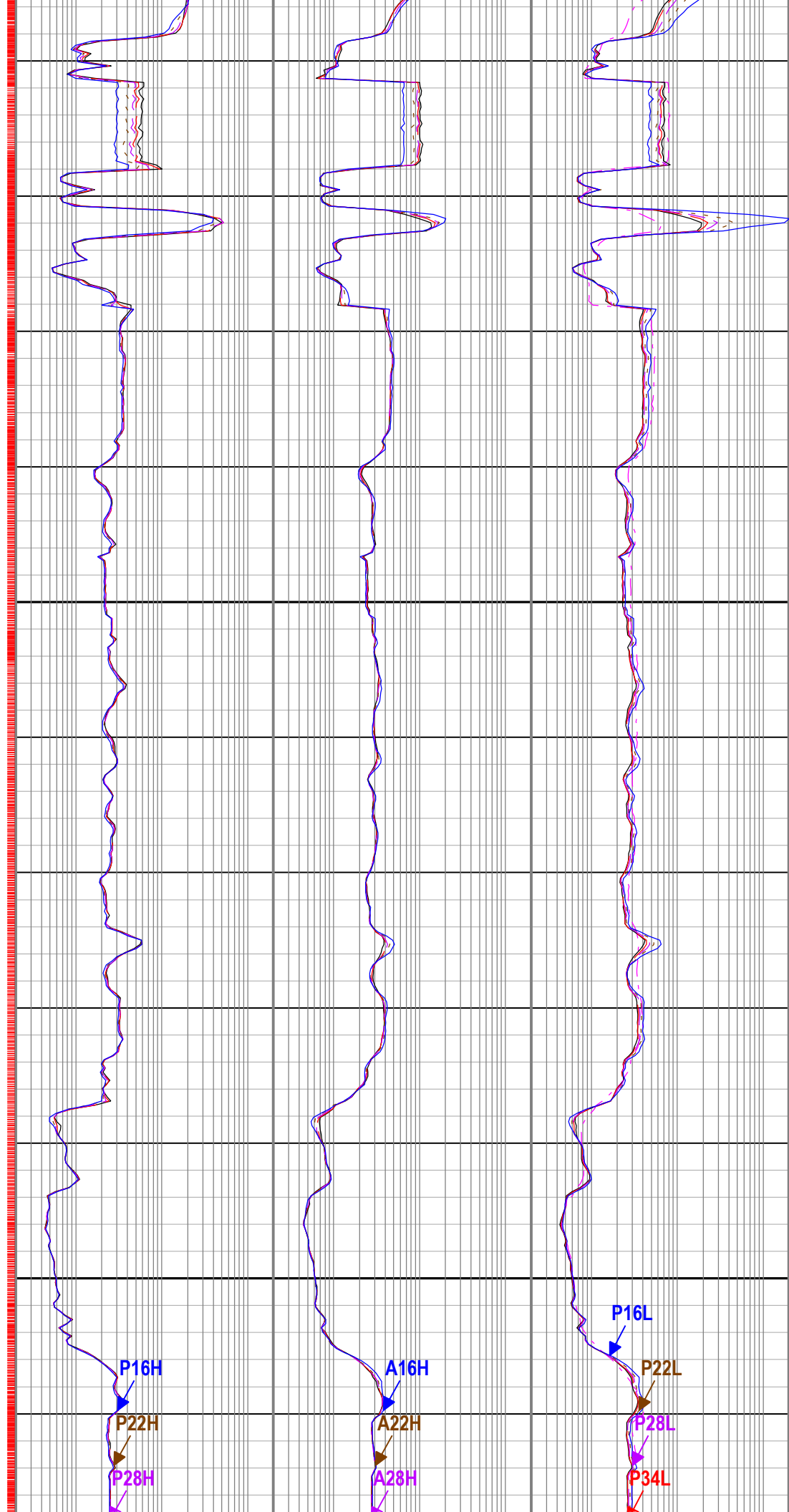
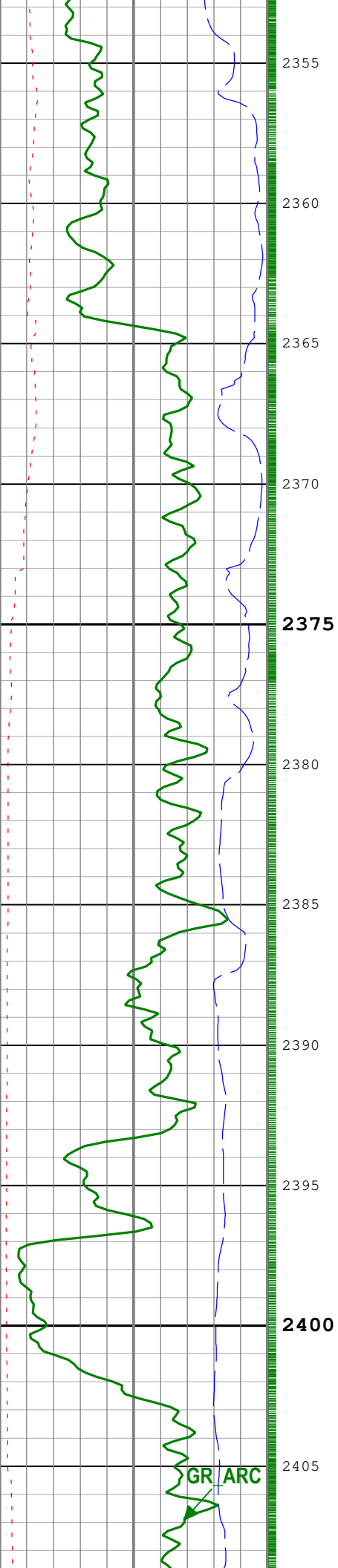


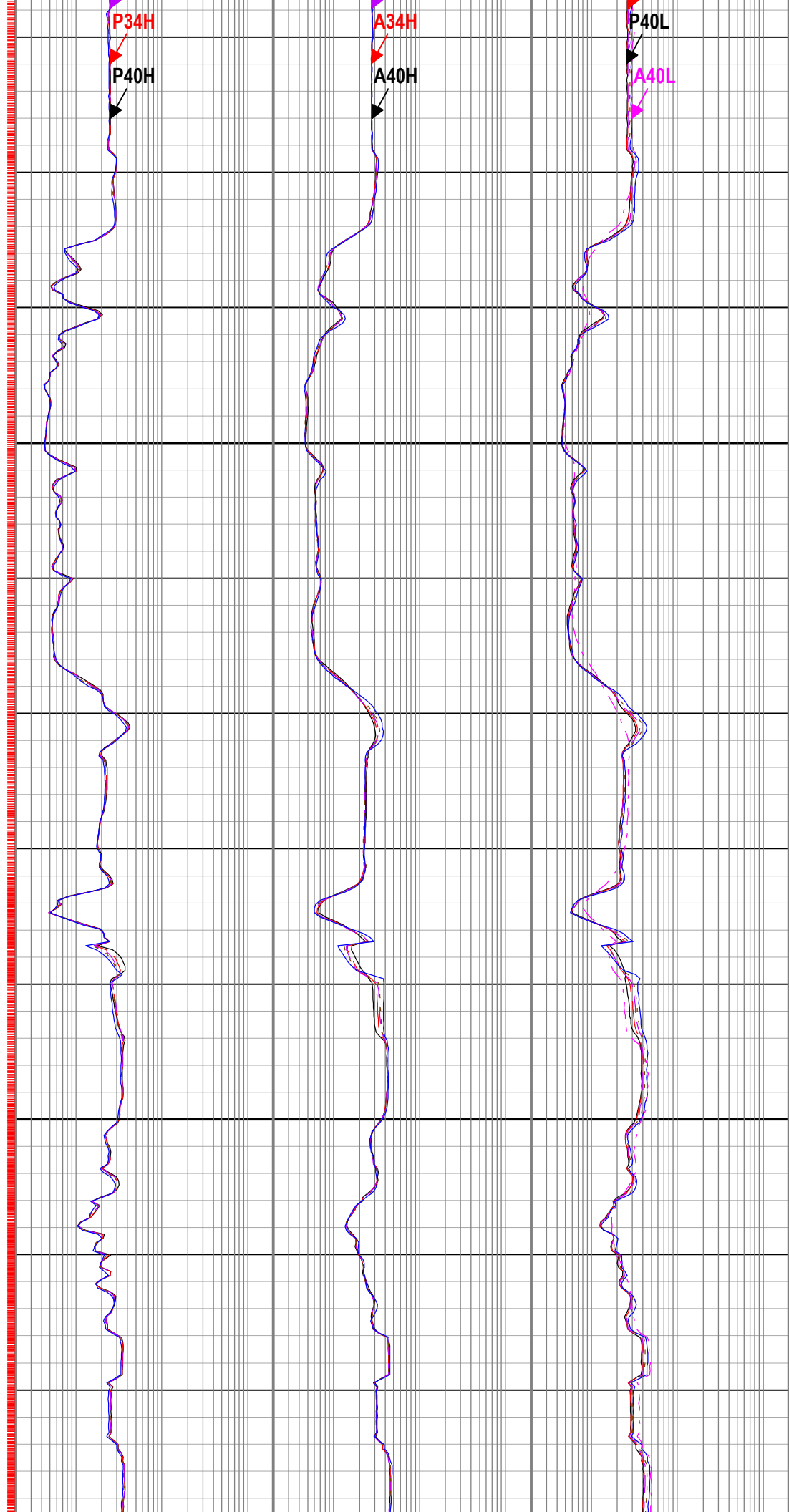
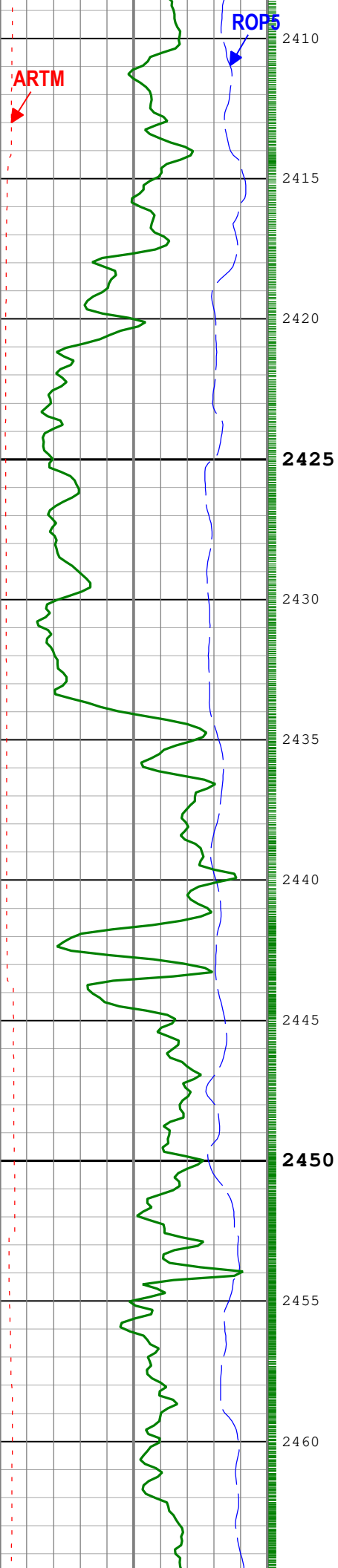


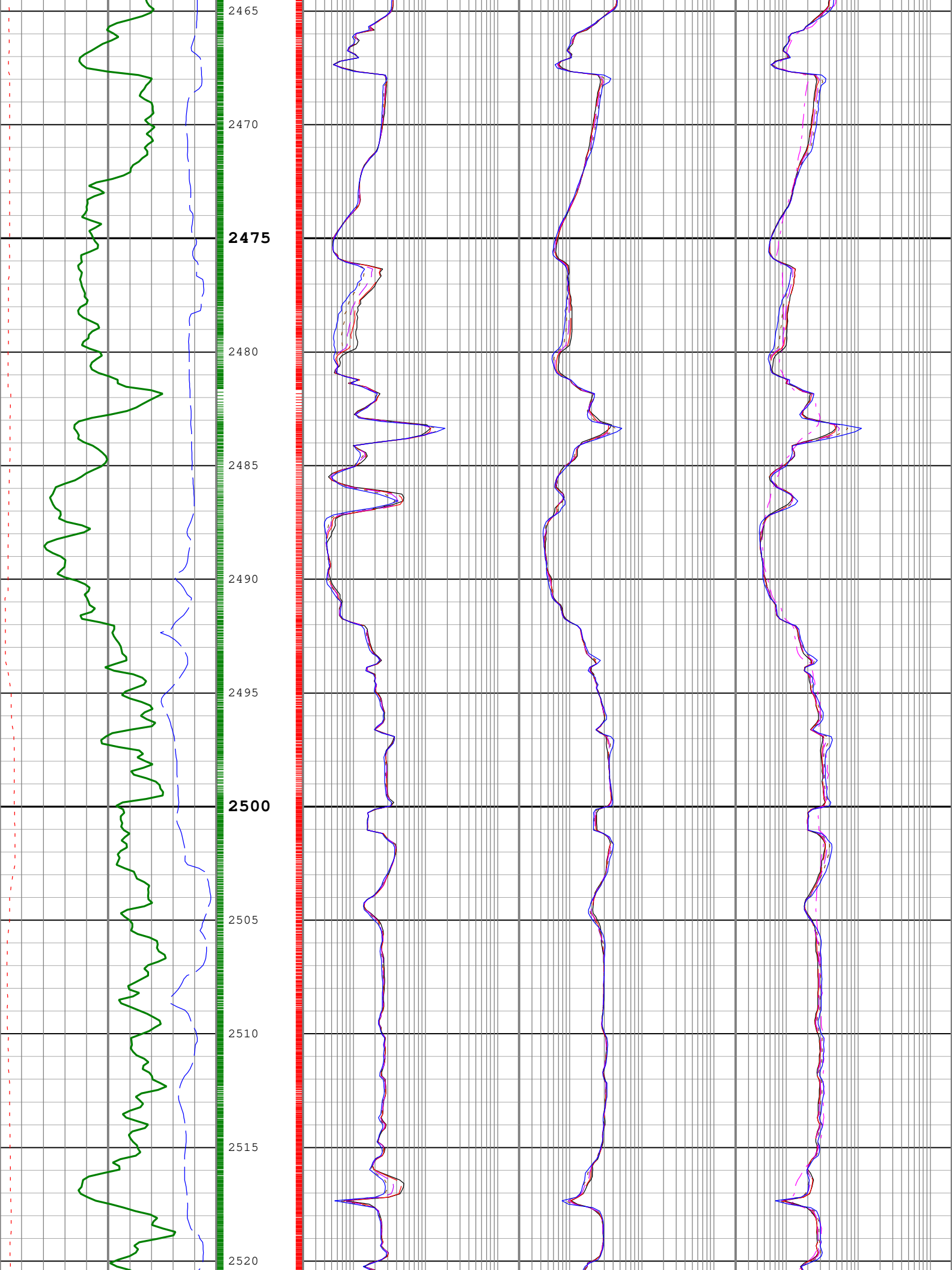


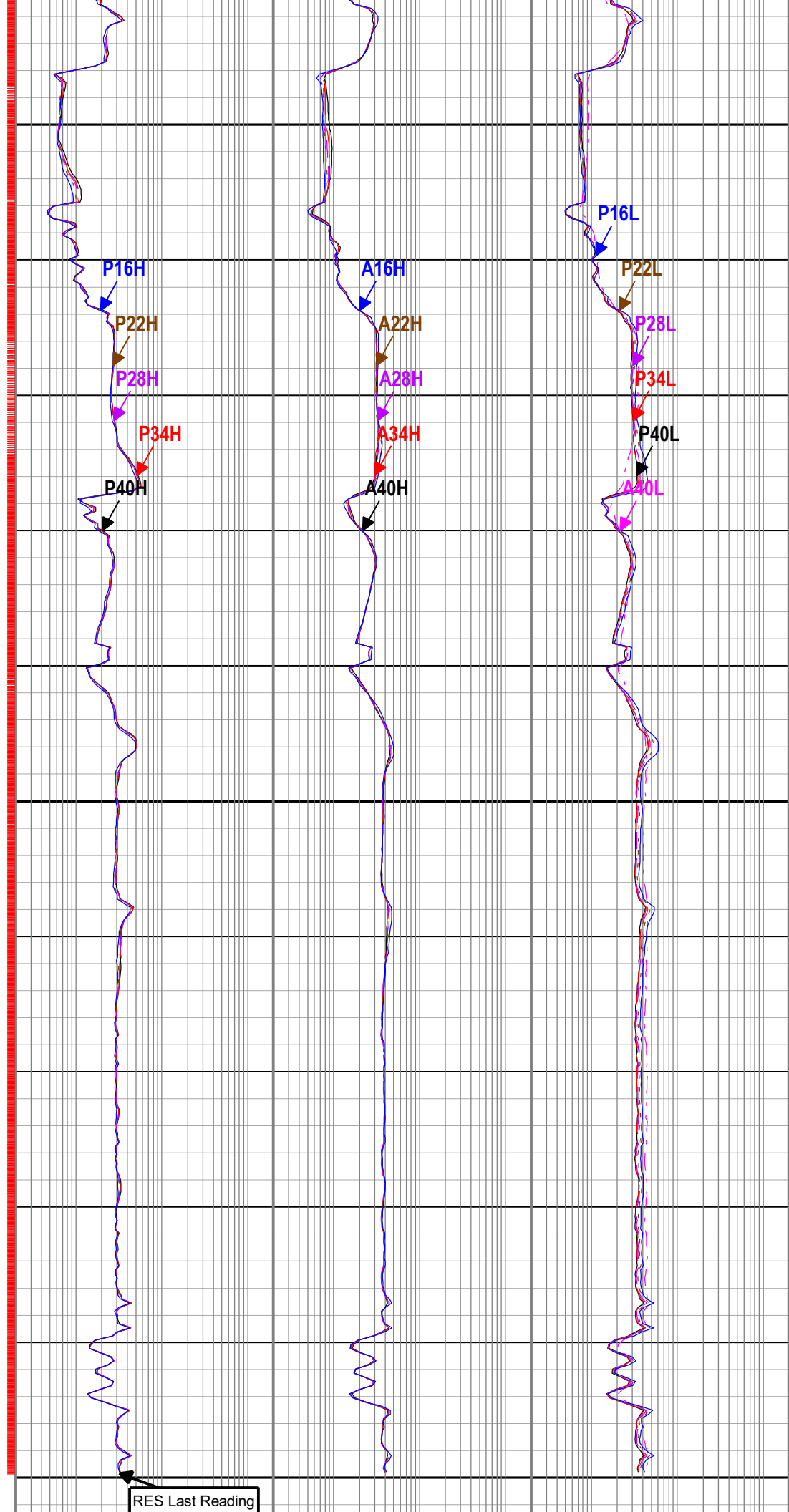
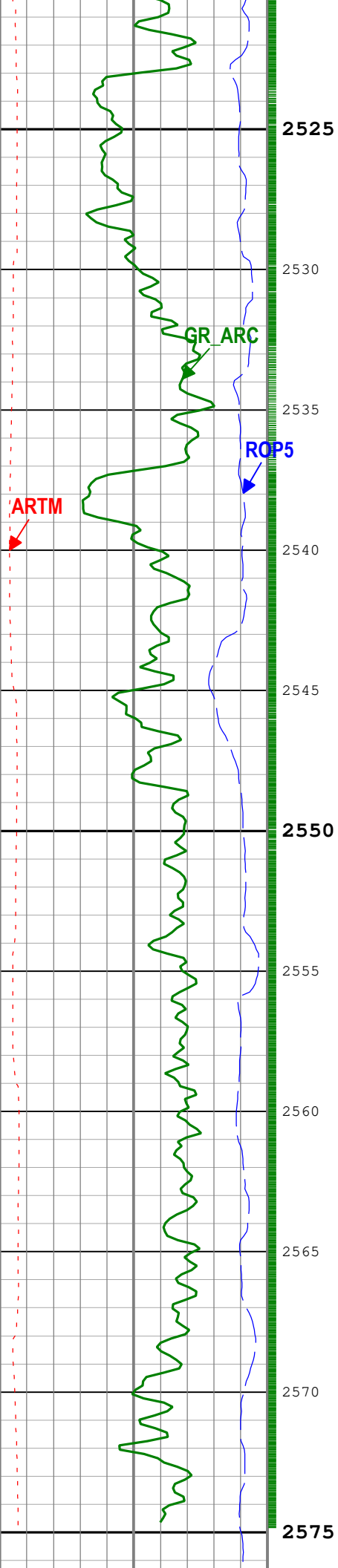


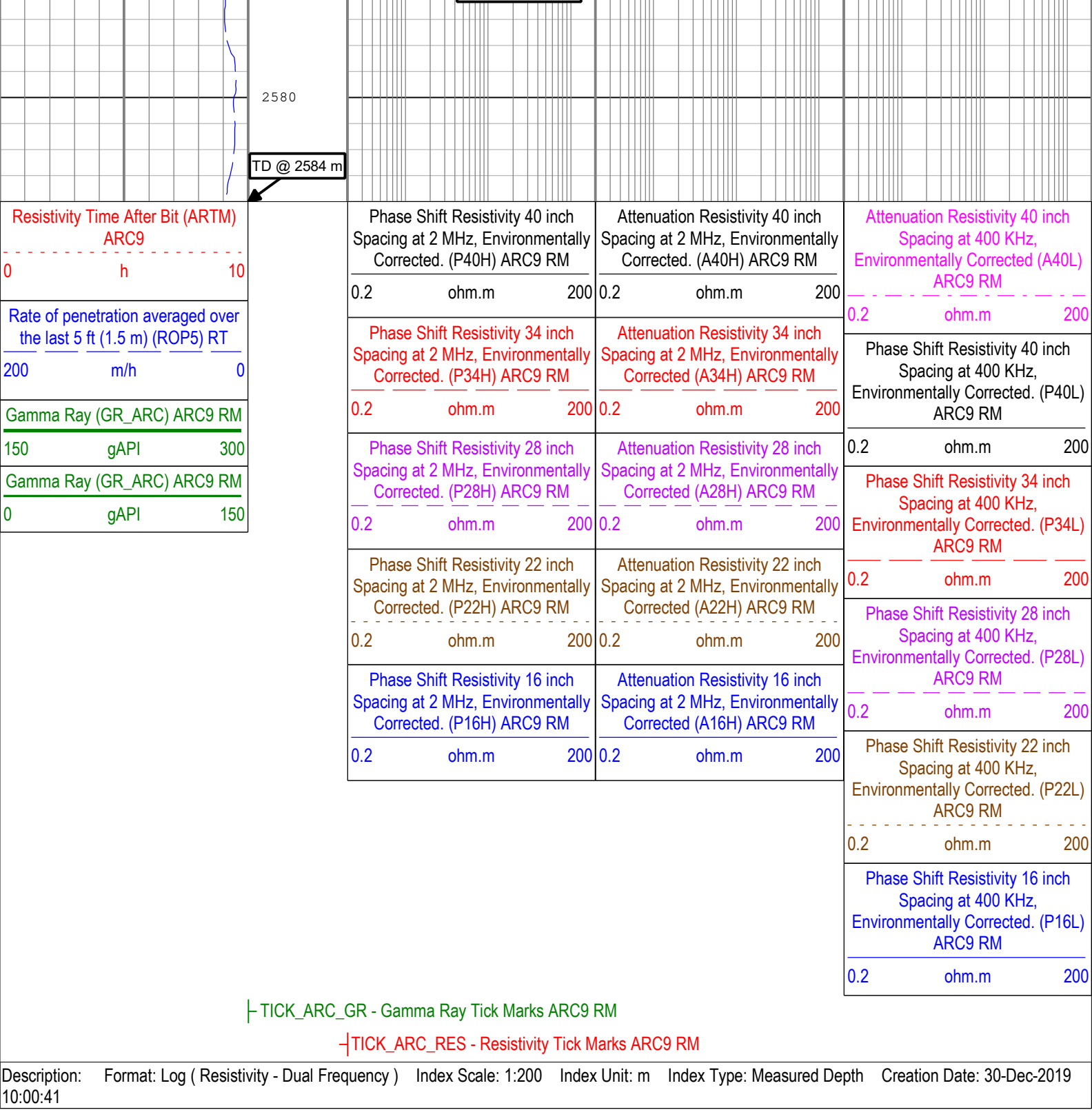












Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:200 Index Unit: m Index Type: Measured Depth Creation Date: 30-Dec-2019 10:00:41

Channel Processing Parameters				
Run 8: Parameters				
Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	ARC9	NO_TX_FAILED	
BHK	Drilling Fluid Potassium Concentration	Borehole	Time Zoned	%
BS	Bit Size	DNMSESSION	Depth Zoned	in
DFD	Drilling Fluid Density	Borehole	Time Zoned	g/cm3
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	
HIGH_BLEND	High Resistivity Threshold for Blending	ARC9	2	ohm.m
LOW_BLEND	Low Resistivity Threshold for Blending	ARC9	1	ohm.m
RMS	Resistivity of Mud Sample	Borehole	0.07	ohm.m

Depth Zone Parameters

Parameter	Value	Start (m)	Stop (m)
BS	17.5	1856	1904
BS	12.25	1904	2584

All depth are actual.

Time Zone Parameters

Parameter	Value	Start Time	Stop Time	Start Depth (m)	Stop Depth (m)
BHK	7.12	23-Dec-2019 06:55:46	24-Dec-2019 03:30:00	11.43	2010.59
BHK	6.9	24-Dec-2019 03:30:00	25-Dec-2019 13:49:45	2010.59	2583.94
DFD	1.25	23-Dec-2019 06:55:46	24-Dec-2019 15:19:00	11.43	2300
DFD	1.26	24-Dec-2019 15:19:00	25-Dec-2019 13:49:45	2300	2583.94

All depth are at tool zero.

Tool Control Parameters

Calibration Report

ARC9 (Array Resistivity Compensated 900) Calibration - Run 8

Primary Equipment :		
High Pressure without AIM Receiver	AREA	9139

RESAIRCAL - Resistivity: Air

Master (Time Frame File):		14:29:45 27-Nov-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Attenuation T1 at 2 MHz	dB	Master	8.500	6.500	8.526	10.500	
Attenuation T2 at 2 MHz	dB	Master	6.500	4.500	6.012	8.500	
Attenuation T3 at 2 MHz	dB	Master	4.500	2.500	5.305	6.500	
Attenuation T4 at 2 MHz	dB	Master	4.600	2.600	3.995	6.600	
Attenuation T5 at 2 MHz	dB	Master	3.600	1.600	3.920	5.600	
Phase Shift T1 at 2 MHz	deg	Master	0.100	-3.900	1.422	4.100	
Phase Shift T2 at 2 MHz	deg	Master	0.100	-3.900	-1.320	4.100	
Phase Shift T3 at 2 MHz	deg	Master	0.100	-3.900	1.370	4.100	
Phase Shift T4 at 2 MHz	deg	Master	0.100	-3.900	-1.331	4.100	
Phase Shift T5 at 2 MHz	deg	Master	0.100	-3.900	1.348	4.100	
Attenuation T1 at 400 KHz	dB	Master	8.500	6.500	8.514	10.500	
Attenuation T2 at 400 KHz	dB	Master	6.500	4.500	6.029	8.500	
Attenuation T3 at 400 KHz	dB	Master	4.500	2.500	5.288	6.500	
Attenuation T4 at 400 KHz	dB	Master	4.600	2.600	4.009	6.600	
Attenuation T5 at 400 KHz	dB	Master	3.600	1.600	3.909	5.600	
Phase Shift T1 at 400 KHz	deg	Master	0.100	-3.900	0.006	4.100	
Phase Shift T2 at 400 KHz	deg	Master	0.100	-3.900	-0.110	4.100	
Phase Shift T3 at 400 KHz	deg	Master	0.100	-3.900	0.046	4.100	
Phase Shift T4 at 400 KHz	deg	Master	0.100	-3.900	-0.113	4.100	
Phase Shift T5 at 400 KHz	deg	Master	0.100	-3.900	0.024	4.100	

GRGAIN - Gamma Ray: Blanket

Master (Time Frame File):		06:57:38 15-Nov-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Gamma Ray Calibration Gain		Master	1.000	0.640	0.979	1.200	

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway



Schlumberger

VISION Resistivity
Dual Frequency
Recorded Mode